HR Analytics – Employee Attrition Prediction

This project aims to predict employee attrition using machine learning techniques and simulate real-world data challenges by introducing missing values. Orange Data Mining tool was used for model building and evaluation. The workflow includes data preprocessing, model training, and performance analysis using confusion matrices and evaluation metrics.

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1. Introduction

Employee attrition is a critical issue in HR management. Predicting attrition can help organizations retain valuable talent. In this project, a dataset containing HR analytics was used, with intentional missing values introduced to simulate real-world data issues.

We removed 5000 random cells from the original clean dataset to represent data imperfections. These missing values were handled using appropriate imputation techniques, and machine learning models were trained to predict whether an employee is likely to leave or not.

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2. Objective

- To analyze HR data and build predictive models for employee attrition.
- To evaluate and compare the performance of different machine learning algorithms.
- To apply models on unseen data with missing target values and predict attrition.

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3. Methodology

3.1 Tools Used

- Orange Data Mining
- Python (for preprocessing if needed)
- Microsoft Word (for reporting)

3.2 Dataset Handling

- Introduced 5000 missing cells randomly in the dataset.
- Split into two sets:
  - Modeling Dataset (with target column)
  - Prediction Dataset (target column removed)

3.3 Preprocessing

- Missing values handled using statistical imputation (mean/mode).
- Normalized data for model fairness.

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4. Model Development

Models Used:

- Logistic Regression
- Random Forest
- Decision Tree

Each model was trained and evaluated on the modeling dataset using accuracy, precision, recall, F1-score, and MCC.

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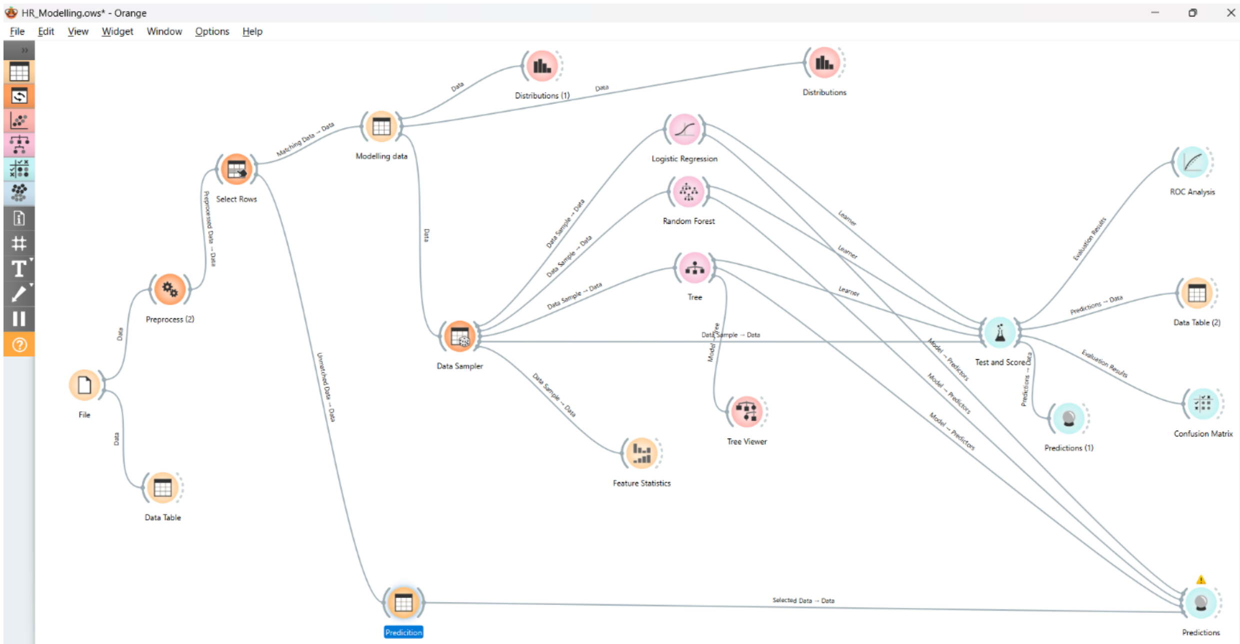
5. Workflow Diagram



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6. Model Evaluation and Results

6.1 Logistic Regression

- True Negatives: 855
- False Positives: 21
- False Negatives: 79
- True Positives: 88
- Accuracy: 90.4%
- Precision: 89.8%
- Recall: 90.4%
- F1 Score: 89.6%
- MCC: 0.603

6.2 Random Forest

- True Negatives: 875
- False Positives: 1
- False Negatives: 42
- True Positives: 125
- Accuracy: 95.9%
- Precision: 96.0%
- Recall: 95.9%
- F1 Score: 95.6%
- MCC: 0.841

6.3 Decision Tree

- True Negatives: 871
- False Positives: 5
- False Negatives: 29
- True Positives: 138
- Accuracy: 96.7%
- Precision: 96.7%
- Recall: 96.7%
- F1 Score: 96.6%
- MCC: 0.875

6.4 Model Comparison Table

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.890	0.904	0.896	0.898	0.904	0.603
Random Forest	0.998	0.959	0.956	0.960	0.959	0.841
Tree	0.989	0.967	0.966	0.967	0.967	0.875

Decision Tree outperformed other models with the highest MCC and accuracy.

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7. Prediction on Unseen Data

Final trained models were used to predict attrition on rows with missing target values. Results were collected using the Predictions widget in Orange.

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8. Orange Widgets Used

Data Import	File, Data Table
Preprocessing	Select Rows, Preprocess, Data Sampler
Modeling	Logistic Regression, Random Forest, Tree
Evaluation	Test & Score, Confusion Matrix, ROC Analysis

Visualization	Tree Viewer, Feature Statistics
Output	Predictions, Data Table

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9. Conclusion

This project demonstrated the effectiveness of machine learning in predicting employee attrition. Simulating missing data helped mimic real-world scenarios, and evaluation showed Decision Tree as the most effective model. These insights can help HR departments make data-informed retention strategies.